Algebra 3 — Factorising · Fractions · Equations with fractions · Simultaneous equations

No calculator. Marks in brackets. ★ = harder question. Check every solution in the original equation(s).

1 Factorise completely: [2]

a $5x^2 + 10x$ **b** $18a^2b - 12ab^2$

> **a** HCF $5x$: $5x(x+2)$. **b** HCF $6ab$: $6ab(3a - 2b)$. Check by expanding.

2 Simplify: [4]

a $\dfrac{x^2 - 3x}{3x}$ **b** $\dfrac{a - ab}{b - b^2}$

> Factorise top and bottom, then cancel the common factor. **a** $\dfrac{x(x-3)}{3x} = \dfrac{x-3}{3}$. **b** $\dfrac{a(1-b)}{b(1-b)} = \dfrac{a}{b}$.

3 Solve: [8]

a $\dfrac{2(x-5)}{3} = -4$ **b** $\dfrac{x-2}{4} = \dfrac{x+2}{5}$ **c** $\dfrac{x}{2} - \dfrac{x}{6} = 3$ **d** $\dfrac{6}{x} = -\dfrac{2}{5}$

> **a** Multiply by 3: $2(x-5) = -12 \Rightarrow x - 5 = -6 \Rightarrow x = -1$.
> **b** Multiply by 20: $5(x-2) = 4(x+2) \Rightarrow 5x - 10 = 4x + 8 \Rightarrow x = 18$.
> **c** Multiply by 6: $3x - x = 18 \Rightarrow 2x = 18 \Rightarrow x = 9$.
> **d** Multiply by $5x$: $30 = -2x \Rightarrow x = -15$. Check: $6 \div (-15) = -0.4 = -\frac{2}{5}$ ✓

4 Solve these simultaneous equations. [6]

a $y = 2x - 5$ and $3x + y = 10$

b $3x + 2y = 12$ and $5x - 2y = 4$

> **a** *Substitution:* put $y = 2x - 5$ into the second:
> $3x + 2x - 5 = 10 \Rightarrow 5x = 15 \Rightarrow x = 3$, then $y = 2(3) - 5 = 1$. Check: $9 + 1 = 10$ ✓
>
> **b** *Elimination:* the y terms are $+2y$ and $-2y$, so add: $8x = 16 \Rightarrow x = 2$. Then $6 + 2y = 12 \Rightarrow y = 3$. Check in the second: $10 - 6 = 4$ ✓

5 At a market, 3 oranges and 2 mangoes cost €3.90, while 2 oranges and 5 mangoes cost €5.90. Find the cost of one orange and of one mango. [5]

> Orange x, mango y: $3x + 2y = 3.90$ (1), $2x + 5y = 5.90$ (2). Eliminate y: (1)$\times 5$: $15x + 10y = 19.50$; (2)$\times 2$: $4x + 10y = 11.80$. Subtract: $11x = 7.70 \Rightarrow x = 0.70$. Then $2.10 + 2y = 3.90 \Rightarrow y = 0.90$. **Orange €0.70, mango €0.90.** Check (2): $1.40 + 4.50 = 5.90$ ✓

6 Solve $\dfrac{x+1}{2} - \dfrac{2x-1}{3} = 1$ [3]

> Multiply by 6:
> $3(x+1) - 2(2x-1) = 6 \Rightarrow 3x + 3 - 4x + 2 = 6 \Rightarrow -x + 5 = 6 \Rightarrow x = -1$.
> Check: $0 - (-1) = 1$ ✓

7 The sum of two numbers is 27 and their difference is 5. Write two equations and solve them to find the numbers. [4]

> $x + y = 27$ and $x - y = 5$. Add: $2x = 32 \Rightarrow x = 16$; then $y = 11$. Check: $16 - 11 = 5$ ✓

Total: 32 marks

Check: clear fractions by multiplying by the lowest common denominator · elimination — match a coefficient, then add or subtract · substitute the pair back into *both* equations.